Multimodal Machine Learning for Alzheimer's Disease Classification Using Clinical Data and MRI Scans from OASIS

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Abstract

This project develops a multimodal machine learning approach to classify Alzheimer's disease using both clinical and imaging data. The study evaluates whether combining clinical and demographic variables with MRI-derived features improves classification performance compared to using either modality alone. To address this, MRI images were processed using a convolutional neural network (CNN) to extract feature representations, which then was combined with clinical patient data. The resulting feature vector was used as input to a multilayer perceptron (MLP) classifier to generate final predictions. Model performance was then compared across clinical-only, MRI-only, and multimodal approaches using standard evaluation metrics.

The clinical-only model achieved the highest overall accuracy and ROC AUC, while the MRI-only model demonstrated weak and unstable performance. The multimodal model achieved the most balanced performance, improving sensitivity to dementia cases compared to the clinical-only model. These findings suggest that MRI-derived features provide complementary information that can enhance Alzheimer's disease classification when integrated with clinical variables.

Introduction

Alzheimer's disease (AD), a neurodegenerative disease and common form of dementia, affects one's ability of thinking, remembering, and reasoning. Approximately, more than 55 million people in the world live with AD or dementia, with this number expected to reach 139 million in the next 50 years (Alzheimer's Disease International). Alzheimer's disease accounts for approximately 60–80% of dementia cases and is among the leading causes of death worldwide, highlighting its growing impact as a major public health concern (Alzheimer's Association, 2024).

Despite extensive research, there is currently no cure for the disease, making early and accurate diagnosis essential for patient care and intervention. Many studies on AD have

utilized neuroimaging techniques such as magnetic resonance imaging (MRI) to detect structural brain changes associated with the disease's progression, including brain atrophy and volume loss (Shi et al., 2025). However, some researchers have highlighted the concern that these measures such as MRIs commonly used in AD research are rather nonspecific indicators of damage to the brain that could be from a variety of other causes. (Jack et al., 2018).

Recent advances in machine learning have enabled the development of predictive models that can analyze complex biomedical data for disease classification. In particular, deep learning approaches have shown promise in identifying Alzheimer's-related patterns from neuroimaging data (Vieira et al., 2017). It is important to note however that many existing studies rely on a single data modality, such as imaging alone, which may limit predictive performance. Integrating multiple data types, including both imaging and clinical variables, may provide a more comprehensive representation of disease status.

This study implemented machine learning approaches to evaluate the effectiveness of multimodal data integration for Alzheimer's disease classification. Specifically, the study investigated whether combining clinical and demographic variables from the OASIS-1 dataset with MRI-based features derived from OASIS MRI slices improves classification performance compared to using either clinical data or MRI data alone. It was hypothesized that the multimodal model would perform best, as clinical variables and MRI scans provide complementary information about disease status.

Methodology

The data for this study was obtained from the Open Access Series of Imaging Studies (OASIS), collected by Washington University School of Medicine. Specifically, the OASIS-1 dataset consists of a cross-sectional collection of 436 subjects aged 18 to 96. The OASIS-1 demographic and clinical dataset was used for the structured (tabular) component of this study, while MRI scans derived from OASIS-1 and made available as preprocessed image slices on Kaggle were used for the imaging component. The MRI images were organized into labeled folders ('Non Demented', 'Very Mild Dementia', 'Mild Dementia', and 'Moderate Dementia'), with patient identifiers embedded in the filenames, allowing for alignment with the clinical dataset.

The first step of the analysis involved loading and preprocessing both the clinical and MRI data. Exploratory data analysis (EDA) was conducted to examine class distributions and assess data quality. The clinical dataset was filtered to include only relevant variables

for this study, including age, sex, normalized whole brain volume (nWBV), and education (Educ). The Clinical Dementia Rating (CDR) was used as the target variable, where CDR = 0 indicates non-demented individuals and CDR > 0 indicates demented or Alzheimer's-related cases. This formulation allowed the problem to be treated as a binary classification task.

To ensure data consistency and model reliability, all rows containing missing values in the selected variables were removed. This step ensured that the models were trained on complete and comparable data without introducing bias from imputation methods. The MRI data was imported directly from Kaggle into Google Colaboratory, and patient IDs extracted from image filenames were used to match MRI scans with corresponding clinical records. Only patients with both complete clinical data and available MRI scans were retained, resulting in a final dataset of matched subjects used across all models.

Model 1: Clinical- Only with Logistic Regression

To establish a baseline for Alzheimer's disease classification, a clinical-only model was implemented using logistic regression. The model utilized the four clinical variables previously mentioned and sex was encoded as a binary variable using one-hot encoding. The dataset was split into training and testing sets using an 80/20 split, with stratification applied to preserve the class distribution of demented and non-demented individuals. Prior to model training, all continuous features were standardized using z-score normalization to ensure that variables were on a comparable scale and to improve model convergence.

A logistic regression classifier was then trained on the standardized training data. Logistic regression models the probability of class membership using a sigmoid function applied to a linear combination of input features, making it well-suited for binary classification problems. The model was trained using default hyperparameters, with a fixed random seed to ensure reproducibility. Model performance was evaluated on the test set using multiple metrics, including accuracy, precision, recall, F1-score, and receiver operating characteristic area under the curve (ROC AUC). A confusion matrix and ROC curve were also generated to visualize classification performance. Additionally, model coefficients were examined to assess the relative importance of each feature, providing insight into the contribution of clinical variables to Alzheimer's disease prediction.

Model 2: MRI-only with Convolutional Neural Network

To evaluate the predictive power of imaging data alone, a convolutional neural network (CNN) was implemented using MRI slices from the OASIS Dataset obtained through Kaggle. Since each patient has multiple MRI slices, a fixed number of slices, in this case N = 5, were selected per patient to ensure consistent input size. Slices were chosen by sorting images based on slice index and selecting evenly spaced samples. All images were converted to grayscale, resized to a fixed resolution of 128 × 128 pixels, and normalized to values between 0 and 1.

The dataset was split at the patient level into training (70%), validation (15%), and testing (15%) sets, ensuring that all slices from a given patient were contained within a single split to prevent data leakage. Although the model was trained on individual slices, predictions were aggregated at the patient level by averaging probabilities across slices to generate a final classification.

The CNN architecture consisted of three convolutional blocks, each containing a convolutional layer followed by a max-pooling layer. Specifically, the network included convolutional layers with 32, 64, and 128 filters, respectively, each using a 3×3 kernel and ReLU activation function. Max-pooling layers with a 2×2 window were applied after each convolutional layer to progressively reduce spatial dimensions and extract higher-level features. Following the convolutional blocks, the feature maps were flattened and passed through a fully connected dense layer with 128 units and ReLU activation, which served as an intermediate feature representation or embedding, A dropout layer with a rate of 0.5 was applied to reduce overfitting. The final output layer consisted of a single neuron with a sigmoid activation function, producing a probability score for binary classification.

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 126, 126, 32)	320
max_pooling2d (MaxPooling2D)	(None, 63, 63, 32)	0
conv2d_1 (Conv2D)	(None, 61, 61, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 64)	0
conv2d_2 (Conv2D)	(None, 28, 28, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 128)	0
flatten (Flatten)	(None, 25088)	0
embedding (Dense)	(None, 128)	3,211,392
dropout (Dropout)	(None, 128)	0
dense (Dense)	(None, 1)	129

Total params: 3,304,193 (12.60 MB)
Trainable params: 3,304,193 (12.60 MB)
Non-trainable params: 0 (0.00 B)

Figure 1: *Model 2 - MRI-only with CNN model architecture*

The model was trained using the Adam optimizer and binary cross-entropy loss function. Early stopping was implemented based on validation loss to prevent overfitting, with training halted if performance did not improve after several epochs. Model performance was evaluated using the same metrics used for Model 1. However, since predictions were probabilistic, a decision threshold of 0.35 was selected to improve sensitivity to dementia cases, and final predictions were generated at the patient level.

Model 3: Multimodal (Clinical + MRI) with Multilayer Perceptron

To leverage both clinical and imaging information, a multimodal model was developed by combining structured clinical variables with MRI-derived feature representations. The same clinical variables used for Model 1 were included, while imaging features were extracted from the previously trained CNN from Model 2. For each patient, MRI feature representations were obtained by passing a fixed number of slices through the trained CNN and extracting the output from the intermediate dense layer. These slices were then averaged to produce a single patient-level MRI feature vector. This approach allowed the model to summarize structural information from multiple slices into a compact representation. The clinical features and MRI embeddings were scaled separately using

standardization to ensure comparable feature ranges. The two feature sets were then concatenated to form a combined multimodal input vector for each patient.

A multilayer perceptron (MLP) was used as the final classifier. The architecture consisted of two fully connected hidden layers with 128 and 64 units, each using ReLU activation functions. Batch normalization and dropout layers (with rates 0.4 and 0.3) were implemented to improve training stability and reduce overfitting. The final output layer consisted of a single neuron with a sigmoid activation function, producing a probability score for binary classification. The model was trained using the Adam optimizer and binary cross-entropy loss function. Class weights were applied during training to address moderate class imbalance and improve sensitivity to dementia cases. Early stopping was implemented based on validation loss to prevent overfitting.

Model: "sequential_1"

Layer (type)	Output Shape	Param #
dense_1 (Dense)	(None, 128)	768
batch_normalization (BatchNormalization)	(None, 128)	512
dropout_1 (Dropout)	(None, 128)	0
dense_2 (Dense)	(None, 64)	8,256
dropout_2 (Dropout)	(None, 64)	0
dense_3 (Dense)	(None, 1)	65

Total params: 9,601 (37.50 KB)
Trainable params: 9,345 (36.50 KB)
Non-trainable params: 256 (1.00 KB)

Figure 2: *Model 3 - Multimodal (Clinical + MRI) with Multilayer Perceptron model architecture*

Model performance was evaluated on a held-out test set using the same metrics as previous models. As with the MRI-only model, a decision threshold of 0.35 was used to generate final predictions, selected to balance sensitivity and specificity of dementia classification.

Finally, a side-by-side comparison of all three models was visualized to highlight the differences in performance across multiple evaluation metrics and to assess the relative strengths and weaknesses of each modeling approach.

Results and Discussion

Exploratory Data Analysis

Exploratory data analysis was conducted to better understand the distribution and characteristics of the dataset prior to model development. The clinical dataset initially consisted of 436 subjects, however after removing rows with missing values and restricting the dataset to patients with corresponding MRI scans, the final cohort of 200 subjects was used for the analysis.

The distribution of dementia status as shown by Figure 3 was moderately imbalanced, with a greater number of non-demented individuals compared to demented individuals. This imbalance was visualized using a bar chart and informed subsequent modeling decisions, including the usage of class weight and threshold tuning in later models. The relatively small sample size and class imbalance reflect common challenges in biomedical datasets and were considered when interpreting model performance.

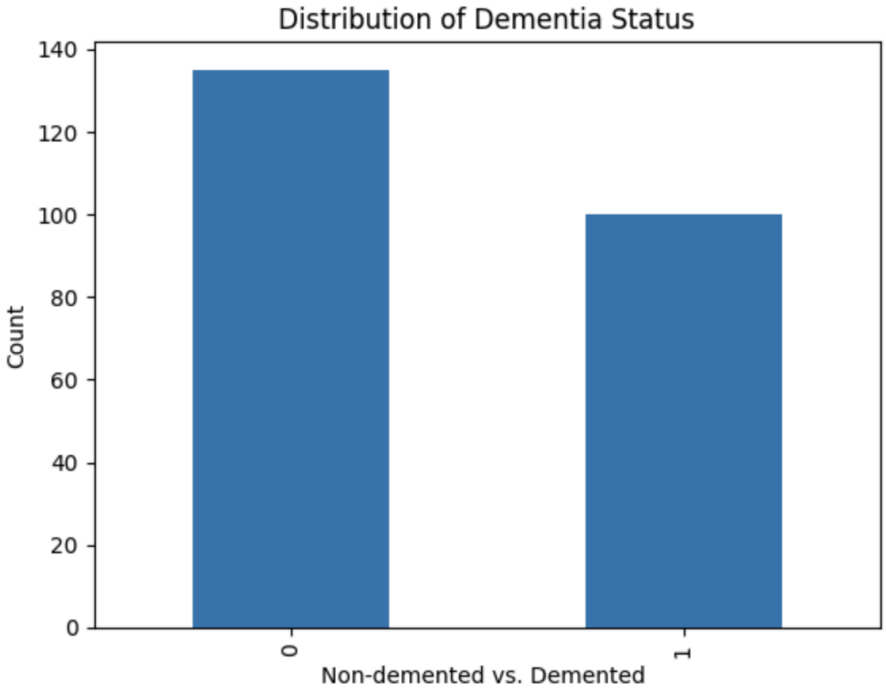
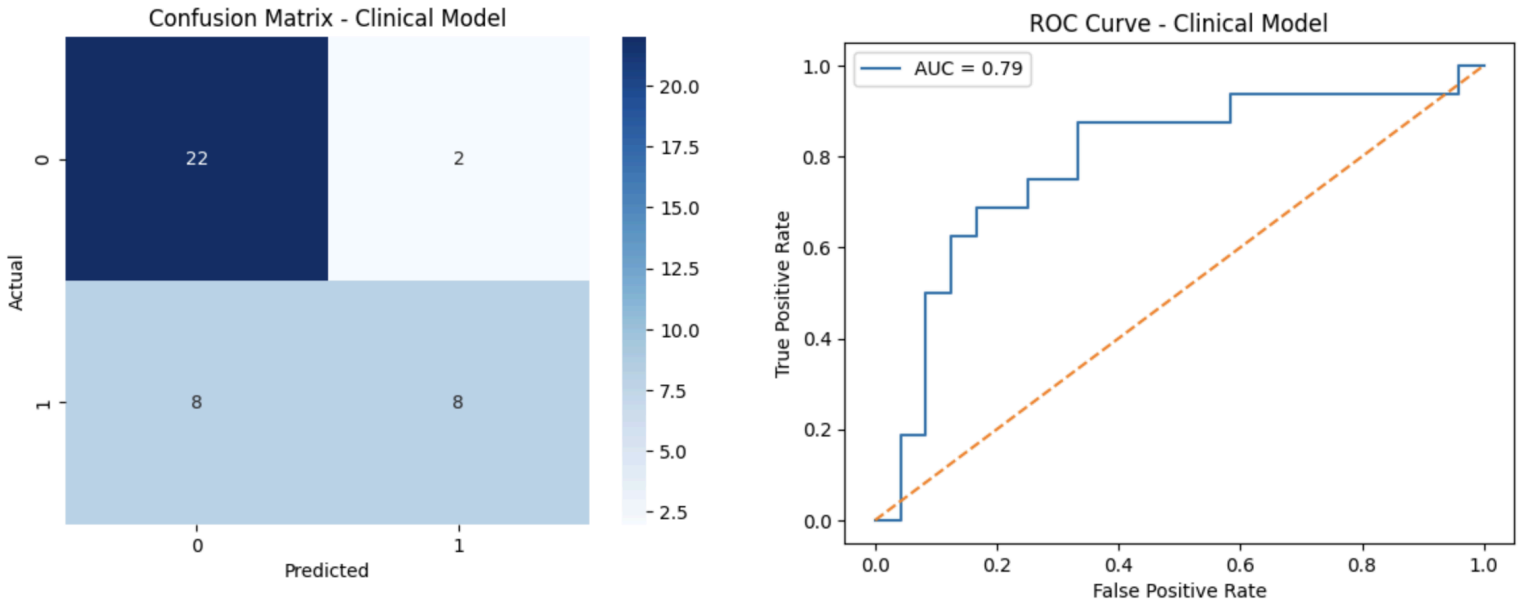


Figure 3: *Distribution of Dementia Status in filtered dataset*

Clinical Model Performance

The clinical-only logistic regression model performed well, achieving the highest overall accuracy among the three models, with an accuracy of 0.75 and a ROC AUC of approximately 0.79. The model demonstrated strong performance in identifying non-demented individuals (class 0), achieving a precision of 0.73, recall of 0.92, and an F1-score of 0.81. However, performance for demented individuals (class 1) was lower, with a precision of 0.80, recall of 0.50, and an F1-score of 0.62.



Accuracy: 0.75

	precision	recall	f1-score	support
0	0.73	0.92	0.81	24
1	0.80	0.50	0.62	16
accuracy			0.75	40
macro avg	0.77	0.71	0.72	40
weighted avg	0.76	0.75	0.74	40

ROC AUC: 0.7864583333333334

Figure 4: *Confusion matrix, ROC curve, and classification metrics for the clinical-only logistic regression model.*

Analysis of feature coefficients revealed that normalized whole brain volume (nWBV) was the most influential predictor, with a strong negative coefficient, indicating that lower brain volume is associated with a higher likelihood of dementia. Education also exhibited a negative relationship with dementia risk, suggesting that higher education may provide a protective effect, consistent with the concept of cognitive reserve. Age and sex showed smaller contributions in comparison.

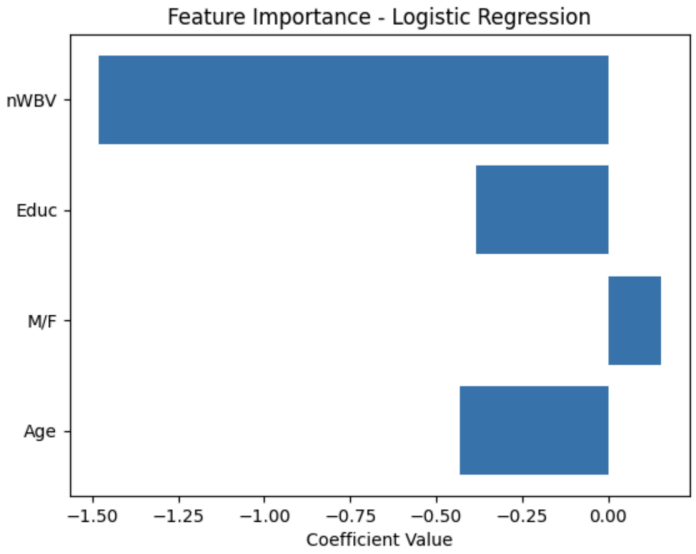


Figure 5: *Feature importance coefficients for the clinical-only logistic regression model.*

While the clinical model demonstrated strong overall performance, its lower recall for dementia cases (0.50), indicates that half of the true dementia cases were misclassified as non-demented, as also reflected in the confusion matrix with 8 false negatives. This limitation highlights the model's reduced sensitivity to detecting Alzheimer's disease, which is particularly important in clinical settings where early identification is critical.

MRI-only Model Performance

To investigate the predictive power of imaging data alone, a CNN was trained using MRI slices from the OASIS dataset. Prior to model training, a fixed number of slices (N = 5) were selected randomly per patient to provide a consistent input structure. An example of these slices is shown in Figure 6, where five representative cross-sectional images from a single patient are displayed. This visualization captures the anatomical structures such as the ventricles and cortical regions, which are known to change in Alzheimer's disease due to brain atrophy, but also highlights that individual slices vary significantly in the amount of visible structural information. Some slices may show more informative regions of the brain, while others contain less relevant information. As a result, relying on a small number of 2D slices may not provide a complete picture of the patient's brain structure. This limitation likely contributes to the difficulty the model faces in distinguishing between demented and non-demented individuals.

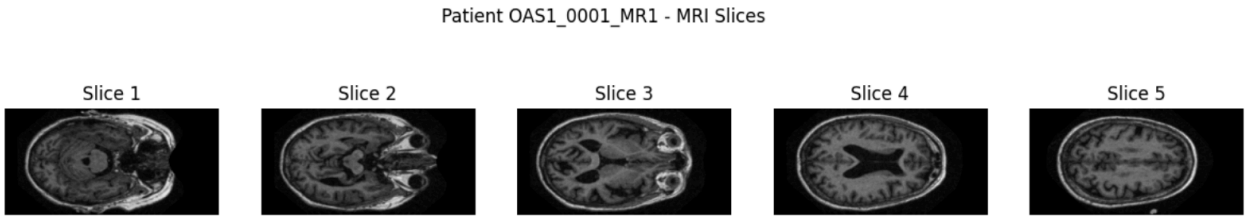


Figure 6: *Representative MRI slices from a single patient used as input to the CNN model.*

During training, the CNN showed slight improvements in performance on the training data, with training accuracy increasing from around 0.59 to 0.67, but validation accuracy decreased from 0.6 to approximately 0.54 with validation loss increasing over time. The divergence between training and validation indicates overfitting, showing that the model fails to generalize to unseen data.

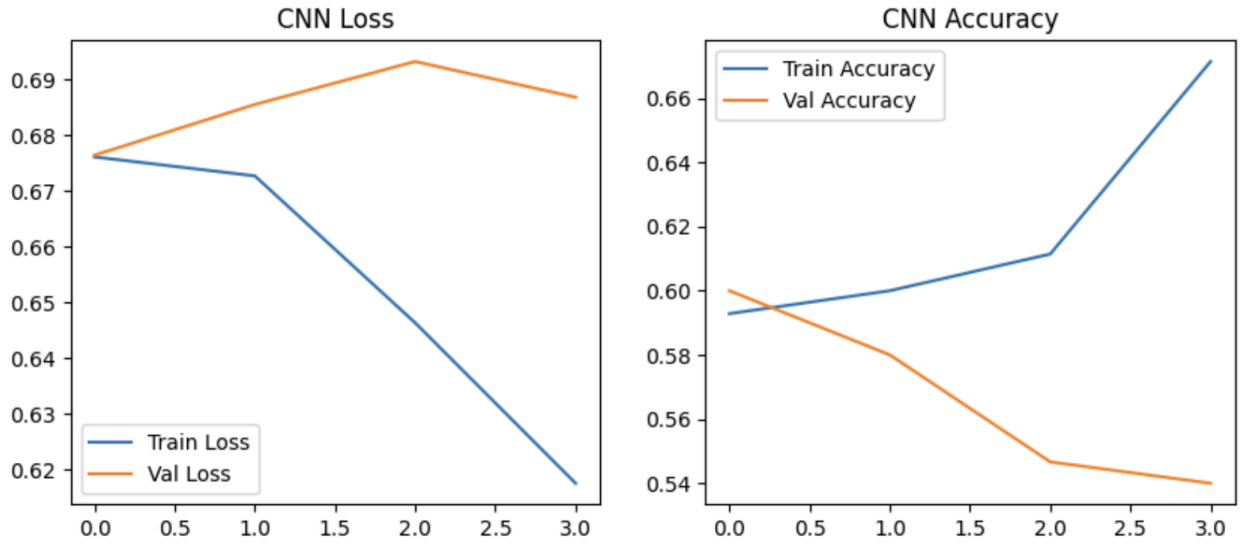
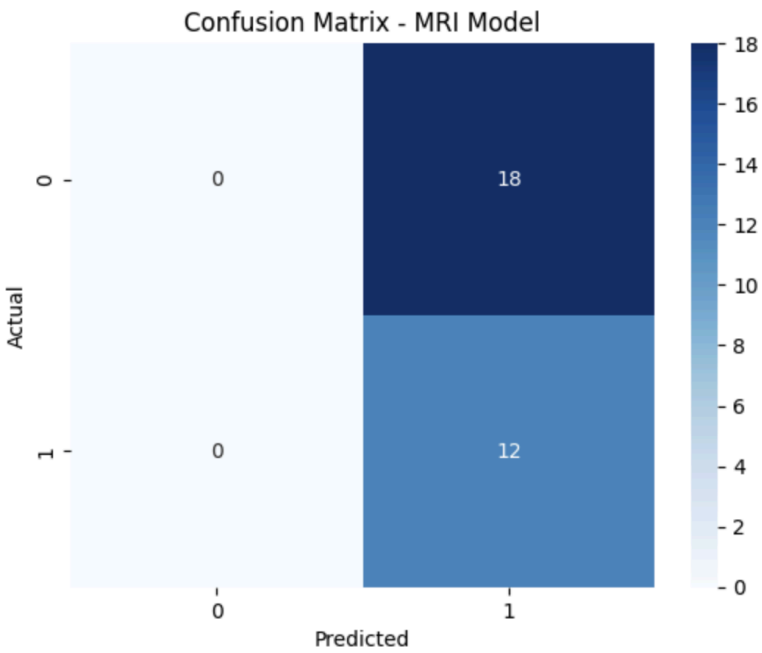
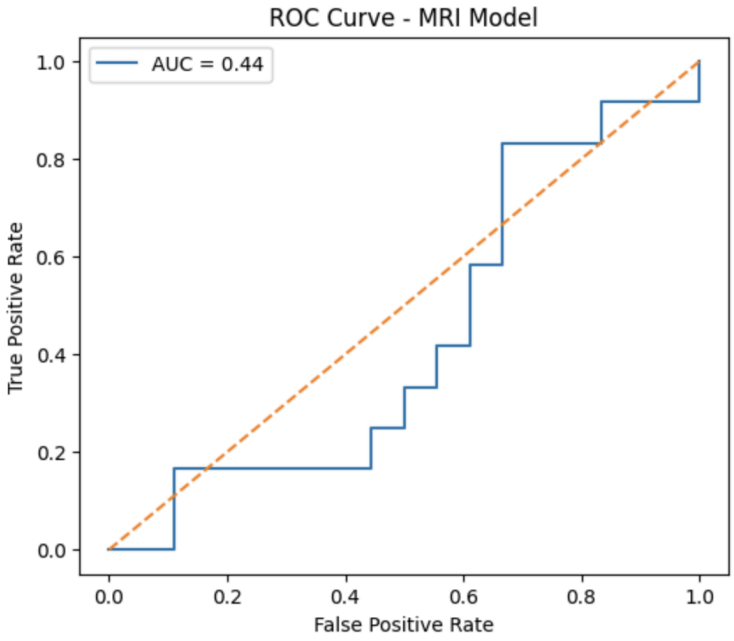


Figure 7: *Training and validation loss and accuracy curves for the MRI-only CNN model.*

On the test set, the model achieved an overall accuracy of 0.4 and a ROC AUC of 0.44, showing poor overall classification performance. The confusion matrix (Figure 8) reveals that the model predicted all test samples as demented (class 1), resulting in 12 true positives, 18 false positives, and 0 true negatives or false negatives. As a consequence, the model achieved a recall of 1 for dementia cases, but achieved a recall of 0 for non-demented individuals, failing to correctly classify any healthy patients. This imbalance indicates that the model is heavily biased toward predicting the positive class and lacks the ability to distinguish between the two classes effectively.

The ROC curve supports this interpretation, with an AUC of 0.44, suggesting that the model performs worse than random guessing in ranking patients by risk. This indicates that the predicted probabilities do not meaningfully separate demented and non-demented individuals.





Accuracy: 0.4

	precision	recall	f1-score	support
0	0.00	0.00	0.00	18
1	0.40	1.00	0.57	12
accuracy			0.40	30
macro avg	0.20	0.50	0.29	30
weighted avg	0.16	0.40	0.23	30

Figure 8: *Confusion matrix, ROC curve, and classification metrics for the MRI-only CNN model.*

These findings highlight the limitations of using MRI slices alone for Alzheimer's classification in this dataset and motivate the use of multimodal approaches that integrate clinical information.

Multimodal Model Performance

The multimodal model combined clinical variables with MRI-derived feature embeddings, resulting in improved overall balance across evaluation metrics. As shown in Figure 9, the model achieved a training accuracy of approximately 0.85-0.88, while validation accuracy stabilized around 0.70-0.73, indicating moderate generalization with some degree of overfitting. Similarly, training loss decreased steadily over time, while validation loss plateaued, further suggesting that the model learned meaningful patterns but did not fully generalize to unseen data.

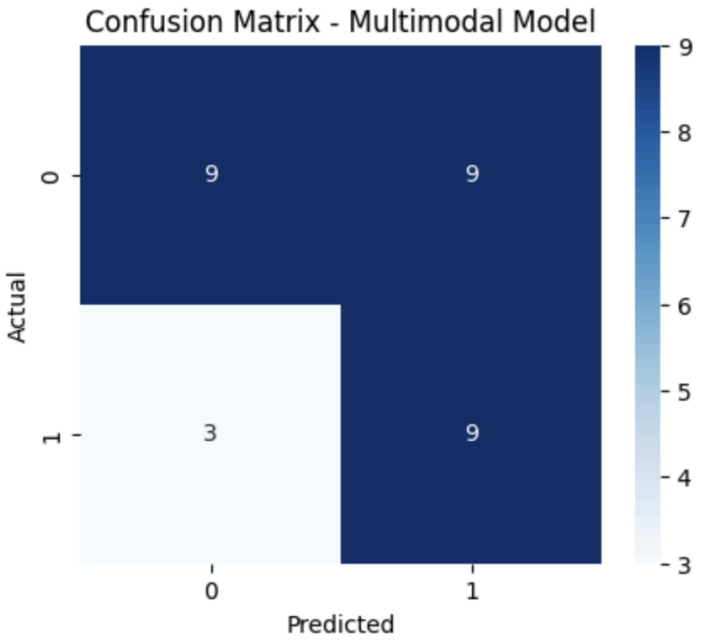


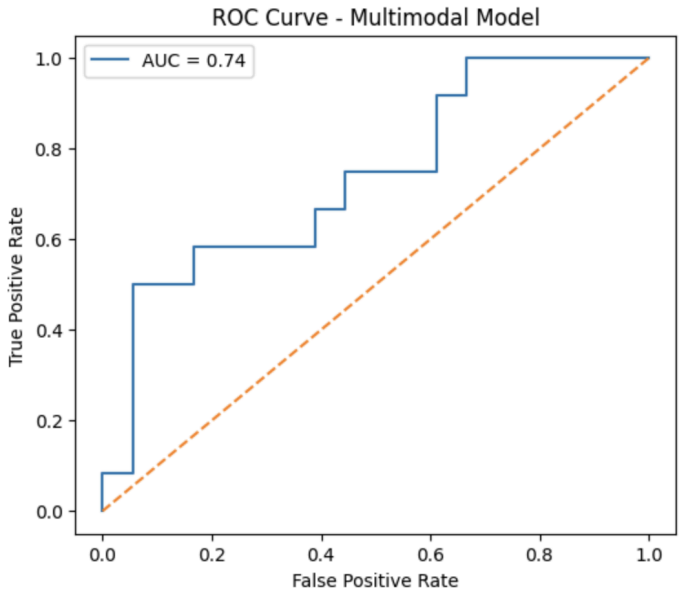
Figure 9: *Training and validation loss and accuracy curves for the multimodal MLP model.*

On the test set, the multimodal model achieved an overall accuracy of 0.60 and a ROC AUC of 0.74, demonstrating improved discriminative ability compared to the MRI-only model and approaching the performance of the clinical model. The confusion matrix (Figure 10) shows that the model correctly identified 9 out of 12 dementia cases (true positives) and 9 out of 18 non-demented cases (true negatives), while misclassifying 3 dementia cases (false negatives) and 9 non-demented cases (false positives).

These results correspond to recall of 0.75 for dementia cases and 0.5 for non-dementia cases, with an F-1 score of 0.6. Compared to the clinical-only model, which achieved a lower recall for dementia (0.50), the multimodal model demonstrates improved

sensitivity to detecting disease. At the same time, compared to the MRI-only model, the multimodal approach provides more stable and balanced performance, avoiding the extreme class bias observed in the MRI model.






```
Accuracy: 0.6
              precision    recall  f1-score   support

           0       0.75      0.50      0.60        18
           1       0.50      0.75      0.60        12

    accuracy                           0.60        30
   macro avg       0.62      0.62      0.60        30
weighted avg       0.65      0.60      0.60        30

ROC AUC: 0.736111111111111
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Figure 10: *Confusion matrix, ROC curve, and classification metrics for the multimodal MLP model.*

The ROC curve further supports these findings, with an AUC of 0.74, indicating that the multimodal model is able to effectively rank patients by risk and capture meaningful distinctions between classes. This suggests that integrating MRI-derived features with clinical variables enhances the model's ability to capture complementary information about Alzheimer's disease.

Model Performance Comparison

To compare the effectiveness of each modeling approach, a side-by-side evaluation of the clinical-only, MRI-only, and multimodal models was performed using accuracy, recall, F1-score, and ROC AUC metrics. A comparison of model performance across key metrics is shown in Figure 11.

	Model	Accuracy	Recall (Class 1)	F1 Score	ROC AUC
0	Clinical	0.75	0.50	0.62	0.79
1	MRI	0.40	1.00	0.57	0.44
2	Multimodal	0.60	0.75	0.60	0.74

Figure 11: *Comparison of model performance between Clinical-only, MRI-only, and Multimodal*

The clinical-only model achieved the highest accuracy and ROC AUC, indicating strong predictive power using structured clinical variables alone. In contrast, the MRI-only model demonstrated weaker performance overall, with lower accuracy and unstable classification behavior, although it exhibited high sensitivity to dementia cases under certain threshold settings. The multimodal model achieved the most balanced performance, integrating the strengths of both clinical and imaging data.

Although the original hypothesis predicted that the multimodal model would outperform both single-modality models across evaluation metrics, the results only partially supported this expectation. While the multimodal model did not exceed the clinical-only model in terms of overall accuracy, it demonstrated improved recall for dementia cases and more stable performance than the MRI-only model, suggesting that MRI-derived features provide complementary information when combined with clinical variables.

Discussion and Implications

The results of this study highlight the differing strengths and limitations of clinical, imaging, and multimodal approaches for Alzheimer's disease classification. As shown above in Figure 11, while the clinical-only model achieved the highest overall accuracy (0.75) and ROC AUC (0.79), it demonstrated limited sensitivity to dementia cases, correctly identifying only 50% of positive cases. In contrast, the MRI-only model

performed poorly overall, with an accuracy of 0.40 and ROC AUC of 0.44, exhibiting unstable behavior, predicting all test samples as demented. Although this resulted in a recall of 1.00 for dementia cases, it completely failed to identify non-demented individuals, indicating a lack of meaningful class separation.

The multimodal model provided the most balanced performance across metrics achieving accuracy of 0.60, recall of 0.75 for dementia cases and ROC AUC of 0.74. While it did not surpass the clinical model in accuracy, it significantly improved sensitivity to detecting dementia. This suggests that MRI-derived features contribute additional information that enhances the model's ability to detect diseases, even if they do not independently provide strong predictive power.

These results demonstrate that clinical variables and MRI data capture complementary aspects of Alzheimer's disease. Clinical features, such as normalized whole brain volume (nWBV) and education level, provide strong predictive signals related to brain atrophy and cognitive reserve (Stern, 2012). In contrast, MRI slices offer structural information about brain anatomy but are limited when used in isolation, particularly when restricted to a small number of 2D slices (Suk and Shen, 2013). The multimodal model integrates these sources, allowing for improved detection of dementia cases while maintaining reasonable overall performance.

Importantly, the findings also highlight that more complex models do not necessarily outperform simpler approaches when data is limited. Small datasets face issues such as data diversity, noise, imbalance, and high dimensionality (Dao et al., 2023) The strong performance of the clinical model suggests that structured clinical data alone can provide substantial predictive value. However, the improved recall observed in the multimodal model indicates that incorporating imaging data can enhance sensitivity, which is important in clinical practice where early detection of Alzheimer's disease is critical.

Several limitations should be considered when interpreting these results. The dataset size was relatively small, particularly after restricting to patients with both clinical and MRI data, which may limit generalizability. Additionally, the MRI data consisted of 2D slices rather than 3D brain volumes, potentially reducing the ability of the CNN to capture complete structural patterns associated with Alzheimer's disease. Prior work has shown that MRI-based prediction improves when leveraging longitudinal 3D volumetric features and region-specific brain measurements, rather than isolated 2D slices, as these approaches better capture neurodegenerative progression over time (Mofrad et al., 2021).

The variability in slice informativeness and the use of a limited number of slices per patient further contributed to the weak performance of the MRI-only model. Further work could address these limitations by incorporating larger datasets, utilizing full 3D MRI volumes, and exploring more advanced multimodal architectures. Additionally, improving feature extraction from imaging data and incorporating longitudinal clinical data may further enhance predictive performance.

Conclusion

Overall, this study demonstrates that while clinical data provides a strong baseline for Alzheimer's classification, multimodal approaches offer a promising direction for improving disease detection by leveraging complementary information from multiple data sources. While advances in deep learning and medical imaging have created opportunities for increasingly complex predictive models, the findings of this study demonstrate that more complex architectures do not necessarily guarantee improved performance, particularly when working with limited biomedical datasets. In this case, structured clinical variables provided strong predictive power, while MRI data alone proved difficult to generalize from when restricted to a limited number of 2D slices. At the same time, the multimodal approach demonstrated that imaging and clinical data capture complementary aspects of Alzheimer's disease. Clinical variables reflect demographic, cognitive, and structural indicators of disease progression, while MRI-derived features provide additional anatomical information that may enhance sensitivity to dementia detection. Although the multimodal model did not outperform the clinical-only model in overall accuracy, it achieved more balanced performance across evaluation metrics and improved sensitivity to positive dementia cases, emphasizing the value of integrating heterogeneous data sources.

More broadly, these findings reinforce the importance of thoughtful model selection, preprocessing, and evaluation strategies in biomedical machine learning research. The study also demonstrates the challenges of applying deep learning models to relatively small datasets, where issues such as overfitting, class imbalance, and limited imaging information can substantially affect performance.

Future work could expand upon this research by incorporating larger datasets, full 3D MRI volumes, longitudinal patient data, and more advanced multimodal architectures. As machine learning methods continue to evolve, multimodal approaches remain a

promising direction for improving Alzheimer's disease detection and supporting more personalized diagnostic strategies in clinical practice.

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